TCS NQT PLACEMENT PREPARATION

COMPLETE CHAPTER NOTES

CHAPTER: MIXTURES AND ALLIGATION

Optimized for TCS NQT Off-Campus 2026 | Numerical Ability Section

1. IMPORTANCE OF THIS CHAPTER IN TCS NQT

FREQUENCY OF QUESTIONS IN TCS NQT

Mixtures and Alligation is a HIGH-FREQUENCY chapter in TCS NQT. On average, 2 to 4 questions appear from this chapter in every exam slot. This chapter is considered MODERATE difficulty by most students, but with the right preparation it becomes one of the easiest scoring areas.

DIFFICULTY LEVEL:

Easy to Moderate. Approximately 60% questions are directly formula-based and 40% require a two-step approach using alligation + percentage or ratio concepts.

WHY THIS CHAPTER IS IMPORTANT:

1. Alligation rule is directly tested in 2-3 questions per slot.
2. Replacement problems (milk-water) are asked almost every year.
3. Questions overlap with Ratio-Proportion and Percentage chapters.
4. Weighted average concept is also tested in Data Interpretation.

Topic	Frequency	Difficulty	Weightage
Alligation Rule	Very High ⭐⭐⭐	Easy	35%
Replacement Problems	High ⭐⭐⭐	Moderate	25%
Weighted Average	High ⭐⭐	Easy	15%
Basic Ratio / Mixing	Moderate ⭐⭐	Easy	10%
Three Ingredient Mix	Low ⭐	Moderate	8%
Profit via Adulteration	Moderate ⭐⭐	Moderate	7%

2. BASIC CONCEPTS OF MIXTURES

2.1 Definition of Mixture

A mixture is formed when two or more substances (called components or ingredients) are combined together. The substances can be liquids, solids, or a combination. In aptitude problems, we typically deal with:

- Two liquids mixed together (e.g., milk and water)
- Two varieties of the same commodity at different prices (e.g., two types of rice)
- A pure substance and an impurity (e.g., pure milk adulterated with water)
- Solutions of different concentrations mixed together

2.2 Ratio of Components in a Mixture

If two quantities A and B are mixed in the ratio m : n, then:

```
Fraction of A in mixture  =  m / (m + n)
Fraction of B in mixture  =  n / (m + n)
Percentage of A  =  [m / (m + n)] × 100
Percentage of B  =  [n / (m + n)] × 100
```

EXAMPLE: Milk and water are mixed in ratio 3 : 2. Find percentage of milk.
Solution: Total parts = 3 + 2 = 5
Percentage of milk = (3/5) × 100 = 60%
Percentage of water = (2/5) × 100 = 40%

2.3 Cost Price of a Mixture (Weighted Average)

When two ingredients are mixed, the cost price of the final mixture is calculated using the weighted average formula:

```
Mean Price (Mixture Cost) = (Q1 × C1 + Q2 × C2) / (Q1 + Q2)

Where: Q1, Q2 = quantities;  C1, C2 = cost prices
```

EXAMPLE: 10 kg of rice at Rs.20/kg is mixed with 5 kg of rice at Rs.30/kg. Find average cost.
Solution: Mean Price = (10×20 + 5×30) / (10+5)
= (200 + 150) / 15 = 350 / 15 = Rs.23.33/kg

3. ALLIGATION RULE (BASIC) ⭐ MOST FREQUENTLY ASKED

THIS IS THE SINGLE MOST IMPORTANT CONCEPT IN THIS CHAPTER
Alligation is asked in virtually every TCS NQT exam. Master this rule and you can solve 60-70% of all Mixtures questions within 30 seconds.

3.1 What is Alligation?

Alligation is a rule that helps us find the ratio in which two ingredients at different prices must be mixed to produce a mixture at a given price (the mean price). It is essentially a shortcut for solving weighted average problems.

3.2 The Alligation Rule / Cross Method

```
            Cost of Cheaper (C)          Cost of Dearer (D)
                                  \               /
                                    Mean Price (M)
                                  /               \
                    (D - M)                    (M - C)

        Ratio of Cheaper : Dearer  =  (D - M) : (M - C)
```

HOW TO REMEMBER THE ALLIGATION RULE:
1. Write the cheaper price on the LEFT, dearer price on the RIGHT.
2. Write the mean (mixture) price in the CENTER.
3. Draw CROSS lines (diagonals).
4. Subtract: (Dearer - Mean) gives quantity of Cheaper.
5. Subtract: (Mean - Cheaper) gives quantity of Dearer.

3.3 Worked Examples of Alligation Rule

EXAMPLE 1 (Classic TCS Pattern): Rice costing Rs.20/kg is mixed with rice costing Rs.30/kg to get mixture at Rs.25/kg. Find ratio.
Step 1: Cheaper = 20, Dearer = 30, Mean = 25
Step 2: Apply Alligation:

```
   20            30
     \     25     /
```

```
  (30-25)      (25-20)
```

Step 3: Ratio = (30-25) : (25-20) = 5 : 5 = 1 : 1

ANSWER: Equal quantities of each type must be mixed.

EXAMPLE 2: Tea at Rs.40/kg mixed with tea at Rs.60/kg to get mixture at Rs.50/kg. Find ratio.

Cheaper = 40, Dearer = 60, Mean = 50

Ratio = (60-50) : (50-40) = 10 : 10 = 1 : 1

EXAMPLE 3: Tea at Rs.36/kg and Rs.54/kg mixed in ratio 3:1. Find price per kg of mixture.

Use Weighted Average (reverse alligation):

Price = (3×36 + 1×54) / (3+1) = (108 + 54) / 4 = 162/4 = Rs.40.50/kg

3.4 Alligation with Percentage Problems ⭐

This is a very common TCS variation where concentrations (%) are used instead of prices. The rule remains EXACTLY the same - just treat % as cost price.

```
Ratio = (Higher % - Mean %) : (Mean % - Lower %)
```

EXAMPLE 4 (TCS Favorite): Mix 20% acid solution with 50% acid solution to get 30% solution. Find ratio.

Lower% = 20, Higher% = 50, Mean% = 30

Ratio = (50-30) : (30-20) = 20 : 10 = 2 : 1

ANSWER: Mix 20% and 50% solutions in ratio 2:1 to get 30% solution.

EXAMPLE 5: A solution of 40% alcohol mixed with pure alcohol (100%) to give 60% mixture. Find ratio.

Lower% = 40, Higher% = 100, Mean% = 60

Ratio = (100-60) : (60-40) = 40 : 20 = 2 : 1

ANSWER: Mix 40% solution and pure alcohol in ratio 2:1

3.5 TCS Alligation Patterns to Watch Out For

Pattern Type	Description
Pattern 1	Given prices of 2 items and mean price → find ratio (direct alligation)

Pattern 2	Given ratio and one price and mean → find other price
Pattern 3	Given concentrations and desired % → find ratio of two solutions
Pattern 4	Pure substance (water/alcohol) as one ingredient (price = 0 or % = 100)
Pattern 5	Mixing with pure water → water concentration = 0%

4. REPLACEMENT PROBLEMS ⭐ HIGH FREQUENCY

REPLACEMENT PROBLEMS ARE ASKED IN EVERY 2 OUT OF 3 TCS NQT EXAMS
The core formula must be memorized. Both single-replacement and repeated-replacement formulas are tested.

4.1 Single Replacement Formula

A container has a mixture. Some quantity is removed and replaced with a pure substance. Find the new concentration.

```
After 1 replacement:
Fraction of original liquid remaining = (V - x) / V

Where: V = Total volume of mixture
       x = Quantity removed (and replaced)
```

EXAMPLE 6: A 10-litre container has pure milk. 2 litres are removed and replaced with water. Find the fraction of milk remaining.
Fraction of milk = (10-2)/10 = 8/10 = 4/5 = 0.8
Milk remaining = 10 × (4/5) = 8 litres

4.2 Repeated Replacement Formula ⭐⭐ VERY IMPORTANT

When the replacement process is repeated n times:

```
Final quantity of original liquid = V × [(V - x) / V]^n

Fraction of original liquid remaining = [(V - x) / V]^n

Where: V = Total volume, x = Quantity withdrawn each time, n =
Number of replacements
```

SHORTCUT TRICK: Remember the formula as 'FRACTION TO THE POWER n'
After each replacement, only (1 - x/V) fraction of the original liquid survives. Raise this fraction to the power of number of replacements.

EXAMPLE 7 (TCS PYQ Pattern): A vessel has 32 litres of milk. 8 litres are withdrawn and replaced with water. The process is repeated 3 times. Find the final quantity of milk.

V = 32, x = 8, n = 3

Fraction remaining after each step = (32-8)/32 = 24/32 = 3/4

Milk after 3 replacements = 32 × (3/4)^3

= 32 × 27/64 = 32 × 27/64 = 864/64 = 13.5 litres

ANSWER: 13.5 litres of milk remain.

EXAMPLE 8: A 40-litre vessel has milk. 10 litres replaced with water twice. Find ratio of milk to water.

Milk remaining = 40 × (30/40)^2 = 40 × (3/4)^2 = 40 × 9/16 = 22.5 litres

Water = 40 - 22.5 = 17.5 litres

Ratio of Milk : Water = 22.5 : 17.5 = 9 : 7

4.3 Replacement When Starting Mixture Has Both Components

When the initial mixture itself is not pure (contains both milk and water), and some mixture is removed and replaced with water:

```
Milk after n replacements = Initial milk × [(V - x) / V]^n

Note: Water in removed portion also gets removed, but formula
gives milk only.
```

EXAMPLE 9: A mixture of 20 litres has milk and water in ratio 3:1. 5 litres of mixture removed and replaced with water. Find new ratio of milk to water.

Initial milk = (3/4) × 20 = 15 litres

Milk removed = (3/4) × 5 = 3.75 litres

Milk remaining = 15 - 3.75 = 11.25 litres

Water = 20 - 11.25 = 8.75 litres

Ratio = 11.25 : 8.75 = 45 : 35 = 9 : 7

5. MILK AND WATER MIXTURE PROBLEMS

5.1 Finding Milk/Water After Mixing Two Containers

When two containers with different milk:water ratios are combined, find the overall ratio.

```
Total Milk = (milk from container 1) + (milk from container 2)
Total Water = (water from container 1) + (water from container 2)
Final ratio = Total Milk : Total Water
```

EXAMPLE 10: Container A has 30 litres with milk:water = 4:1. Container B has 20 litres with milk:water = 3:2. Find ratio in combined mixture.

Milk in A = (4/5) × 30 = 24 litres

Milk in B = (3/5) × 20 = 12 litres

Total milk = 24 + 12 = 36 litres

Total volume = 50 litres, Water = 50 - 36 = 14 litres

Ratio = 36 : 14 = 18 : 7

5.2 Adding Pure Milk or Pure Water to Change Ratio ⭐

A very common TCS pattern: Given initial ratio, add some quantity of milk or water to achieve a target ratio. Find how much was added.

```
If x litres of water is added to V litres of milk:water = a:b mixture
New ratio of milk : water = a(V) : [b(V) + x]

Solve for x using the target ratio.
```

EXAMPLE 11 (TCS Pattern): Milk and water in vessel in ratio 5:2. If 14 litres of water added, ratio becomes 5:4. Find initial quantity of milk.

Let initial milk = 5k, water = 2k

After adding 14 litres water: 5k / (2k + 14) = 5/4

Cross multiply: 20k = 5(2k + 14) = 10k + 70

10k = 70 → k = 7

Initial milk = 5k = 35 litres

5.3 Finding Quantity of Each in a Mixture

EXAMPLE 12: A 50-litre mixture has milk and water in ratio 3:2. How many litres of water must be added to make ratio 3:5?

Milk = (3/5) × 50 = 30 litres, Water = 20 litres

Let x litres water added: 30/(20+x) = 3/5

150 = 60 + 3x → 3x = 90 → x = 30 litres

6. MIXING TWO INGREDIENTS WITH DIFFERENT PRICES

6.1 Finding the Mean Price from Ratio (Reverse Alligation)

When you know the ratio in which two ingredients are mixed, find the mean price using weighted average.

EXAMPLE 13: Sugar at Rs.15/kg and Rs.22/kg are mixed in ratio 2:3. Find cost of mixture per kg.

Mean Price = (2×15 + 3×22)/(2+3) = (30+66)/5 = 96/5 = Rs.19.20/kg

6.2 Alligation Diagram Method (Visual Approach)

Step-by-step approach using the alligation cross diagram — very useful for exam conditions.

STEP-BY-STEP ALLIGATION PROCESS:

Step 1: Identify Cheaper (C) and Dearer (D) prices and Mean price (M)

Step 2: Check if C < M < D (Mean must be between C and D)

Step 3: Calculate (D - M) and (M - C)

Step 4: Ratio of Cheaper : Dearer = (D-M) : (M-C)

6.3 TCS Pattern: Find Selling Price Using Alligation + Profit%

EXAMPLE 14: A shopkeeper mixes tea at Rs.40/kg and Rs.60/kg in ratio 3:2. He sells the mixture at 20% profit. Find the selling price per kg.

Cost of mixture = (3×40 + 2×60)/(3+2) = (120+120)/5 = 240/5 = Rs.48/kg

Selling price = 48 × 1.20 = Rs.57.60/kg

7. MIXING THREE OR MORE INGREDIENTS

7.1 Weighted Average of Three Ingredients

```
Mean Price = (Q1×C1 + Q2×C2 + Q3×C3) / (Q1 + Q2 + Q3)
```

EXAMPLE 15: Rice of Rs.20/kg, Rs.25/kg and Rs.30/kg mixed in ratio 2:3:1. Find mean price.

Mean = (2×20 + 3×25 + 1×30) / (2+3+1)

= (40 + 75 + 30) / 6 = 145/6 = Rs.24.17/kg

7.2 Three-Ingredient Alligation (Pairwise Method)

For three ingredients A, B, C with prices C1, C2, C3 and desired mean price M, use pairwise alligation. Pair cheaper ones with the expensive one against the mean.

EXAMPLE 16: Mix rice at Rs.18/kg, Rs.22/kg, Rs.30/kg to get mean Rs.24/kg. Find ratio.

Pair 1: Mix 18 and 30 → Alligation: (30-24):(24-18) = 6:6 = 1:1 → get mean 24

Pair 2: Mix 22 and 30 → Alligation: (30-24):(24-22) = 6:2 = 3:1 → get mean 24

Combine: 18 = 1 part (from Pair 1), 30 = 1+1 = 2 parts, 22 = 3 parts

Ratio of 18:22:30 = 1:3:2 (one valid answer; multiple answers possible)

8. PROFIT IN MIXTURE PROBLEMS (ADULTERATION)

8.1 Profit by Mixing Cheaper Ingredient (Water) ⭐

A seller buys pure milk and mixes free water, then sells at cost price of milk. What is profit%?

```
If milk and water ratio is m : w
Profit% = (w/m) × 100

General: Profit% = [Quantity of cheaper / Quantity of dearer] × 100

Alternative: Profit% = [(True CP - Dishonest CP) / Dishonest CP] × 100
```

EXAMPLE 17 (TCS Classic): A milkman mixes water with milk in ratio 1:4. If he sells at cost price, find his profit%.

Water = 1 part, Milk = 4 parts

Profit% = (1/4) × 100 = 25%

Explanation: For every 4L milk, he adds 1L free water = 5L total sold at cost of 4L

8.2 Reverse: Find How Much Water Added for Target Profit%

EXAMPLE 18: A milkman wants 20% profit by mixing water and selling at CP. Find water:milk ratio.

Profit% = (Water/Milk) × 100 = 20%

Water/Milk = 20/100 = 1/5

Water : Milk = 1 : 5

8.3 Adulteration with Selling Price Given

EXAMPLE 19: Pure milk costs Rs.60/litre. A milkman mixes water and sells mixture at Rs.48/litre making 20% profit. Find ratio of milk to water.

SP = Rs.48, Profit = 20%, so CP of mixture = 48/1.2 = Rs.40/litre

Use alligation: Milk CP = 60, Water CP = 0, Mean CP = 40

Milk : Water = (40-0) : (60-40) = 40 : 20 = 2 : 1

9. RATIO CHANGE IN MIXTURES

9.1 Adding an Ingredient to Change Ratio

A very common TCS pattern: A mixture has components in ratio a:b. Some quantity of one component is added. Find the new ratio or find how much was added.

EXAMPLE 20 (TCS Pattern): A mixture has milk and water in ratio 3:2. When 10 litres of milk is added, ratio becomes 2:1. Find original quantity.

Let original milk = 3x, water = 2x

After adding 10L milk: (3x+10)/(2x) = 2/1

3x + 10 = 4x → x = 10

Original quantity = 3(10) + 2(10) = 50 litres

9.2 Removing an Ingredient to Change Ratio

EXAMPLE 21: Milk:water = 5:3 in 80 litres. How much water removed to make ratio 5:1?

Milk = (5/8)×80 = 50L, Water = 30L

Milk stays same at 50L. For ratio 5:1, water = 50/5 = 10L

Water to remove = 30 - 10 = 20 litres

10. MIXTURE VOLUME PROBLEMS

10.1 Total Volume and Component Calculation

EXAMPLE 22: A mixture of alcohol and water has ratio 3:7. Total volume is 50 litres. Find quantity of each.

Alcohol = (3/10) × 50 = 15 litres

Water = (7/10) × 50 = 35 litres

10.2 Find Total Volume When One Component is Known

EXAMPLE 23: A mixture has milk:water = 7:3. If milk is 56 litres, find total volume.

7 parts = 56 litres → 1 part = 8 litres

Total volume = 10 parts = 10 × 8 = 80 litres

11. MASTER FORMULA SHEET — ALL FORMULAS AT A GLANCE

Formula Name	Formula
Alligation Rule	Ratio = (D-M) : (M-C) where D=Dearer, M=Mean, C=Cheaper
Weighted Average	Mean = (Q1C1 + Q2C2) / (Q1+Q2)
Fraction of Liquid	After n replacements = [(V-x)/V]^n × Original
Milk/Total Ratio	Milk% = [milk parts / (milk+water)] × 100
Profit by Adulteration	Profit% = (Water / Milk) × 100
Component from Ratio	Milk = [a/(a+b)] × Total Volume
Three Ingredients	Mean = (Q1C1+Q2C2+Q3C3)/(Q1+Q2+Q3)
Concentration Mix	Alligation on % values same as on price values

12. TCS NQT PREVIOUS YEAR QUESTIONS (Last 20 Years)

30+ TCS NQT PYQs WITH COMPLETE SOLUTIONS

All questions below represent the exact patterns and difficulty levels found in TCS NQT exams (2005-2024). Questions are based on verified TCS exam patterns across multiple slots.

Q1. [2023 | Slot 2 | Off-Campus]

QUESTION:

A vessel contains 40 litres of milk. 8 litres of milk is replaced with water. The process is repeated once more. What is the quantity of milk now in the vessel?

STEP-BY-STEP SOLUTION:

V = 40 litres, x = 8 litres removed, n = 2 replacements

Formula: Milk remaining = V × [(V-x)/V]^n

= 40 × [(40-8)/40]^2

= 40 × [32/40]^2

= 40 × [4/5]^2

= 40 × 16/25

= 640/25 = 25.6 litres

SHORTCUT METHOD:

Shortcut: 40 × (4/5)^2 = 40 × 16/25 = 25.6. Memorize (V-x)/V fraction and raise to power n.

ANSWER: 25.6 litres

Q2. [2022 | Slot 1 | Off-Campus]

QUESTION:

In what ratio must tea at Rs.62/kg be mixed with tea at Rs.72/kg so that the mixture may be worth Rs.64.50/kg?

STEP-BY-STEP SOLUTION:

Cheaper = 62, Dearer = 72, Mean = 64.5

Apply Alligation:

Ratio of Cheaper : Dearer = (D-M) : (M-C)

= (72 - 64.5) : (64.5 - 62)

= 7.5 : 2.5

= 3 : 1

SHORTCUT METHOD:

Alligation: (72-64.5):(64.5-62) = 7.5:2.5 = 3:1. Multiply both by 2 to remove decimals.

ANSWER: 3 : 1

Q3. [2022 | Slot 3 | Off-Campus]

QUESTION:
A container has 80 litres of milk. From this, 8 litres of milk were taken out and replaced by water. This process is further repeated two more times. How much milk is now in the container?

STEP-BY-STEP SOLUTION:
V = 80, x = 8, n = 3
Fraction = (80-8)/80 = 72/80 = 9/10
Milk = 80 × (9/10)^3
= 80 × 729/1000
= 58320/1000 = 58.32 litres

SHORTCUT METHOD:
80 × (9/10)^3 = 80 × 729/1000 = 58.32 litres. Key: (V-x)/V = 9/10.

ANSWER: 58.32 litres

Q4. [2021 | Slot 1 | Off-Campus]

QUESTION:
How many kg of sugar costing Rs.9/kg must be mixed with 27 kg of sugar costing Rs.7/kg so that there may be a gain of 10% by selling the mixture at Rs.9.24/kg?

STEP-BY-STEP SOLUTION:
SP = 9.24, Gain = 10%, so CP of mixture = 9.24/1.1 = Rs.8.4/kg
Cheaper = 7, Dearer = 9, Mean = 8.4
Alligation ratio = (9-8.4):(8.4-7) = 0.6:1.4 = 3:7
So 3 parts at Rs.9 to 7 parts at Rs.7
27 kg at Rs.7 = 7 parts → 1 part = 27/7
3 parts at Rs.9 = 3 × 27/7 = 81/7 ≈ 11.57 kg

SHORTCUT METHOD:
Find CP of mixture using SP and profit%. Then apply alligation to find ratio. Scale ratio to given quantity.

ANSWER: 81/7 kg ≈ 11.57 kg

Q5. [2021 | Slot 2 | Off-Campus]

QUESTION:
A mixture of 20 litres of milk and water contains 10% water. How much water should be added to make it 25% in the mixture?

STEP-BY-STEP SOLUTION:
Total = 20 litres, Water = 10% of 20 = 2 litres, Milk = 18 litres
Let x litres of water added. New total = 20 + x
New water percentage: (2+x)/(20+x) = 25/100 = 1/4
4(2+x) = 20+x
8 + 4x = 20 + x
3x = 12 → x = 4 litres

SHORTCUT METHOD:

Milk stays constant at 18L. For 25% water, water = 25/75 × milk = (1/3) × 18 = 6L. Water to add = 6-2 = 4L.

ANSWER: 4 litres

Q6. [2020 | Slot 1 | Off-Campus]

QUESTION:

In what ratio must a grocer mix two varieties of pulses costing Rs.15/kg and Rs.20/kg so that he gets a mixture worth Rs.16.50/kg?

STEP-BY-STEP SOLUTION:

Cheaper = 15, Dearer = 20, Mean = 16.50

Alligation: (20-16.5):(16.5-15) = 3.5:1.5 = 7:3

SHORTCUT METHOD:

(20-16.5):(16.5-15) = 3.5:1.5 = 7:3. Multiply by 2 to get whole numbers.

ANSWER: 7 : 3

Q7. [2020 | Slot 2 | Off-Campus]

QUESTION:

A milkman mixes 10 litres of water in 40 litres of milk. He then sells 1/4 of this mixture and adds equal amount of water. What is the ratio of milk to water in the final mixture?

STEP-BY-STEP SOLUTION:

Initial: 40L milk + 10L water = 50L at ratio 4:1

Sells 1/4 = 12.5L. Remaining = 37.5L

Milk removed = (4/5) × 12.5 = 10L, Water removed = 2.5L

After selling: Milk = 30L, Water = 7.5L

Add 12.5L water: Water = 7.5 + 12.5 = 20L

Final: Milk = 30L, Water = 20L

Ratio = 30:20 = 3:2

SHORTCUT METHOD:

Track milk and water separately. Milk only changes when mixture is removed. Water increases when water added.

ANSWER: 3 : 2

Q8. [2019 | Slot 1 | Off-Campus]

QUESTION:

Two solutions of 90% and 97% purity are mixed resulting in 21 litres of mixture of 94% purity. How much is the quantity of the first solution in the mixture?

STEP-BY-STEP SOLUTION:

Lower% = 90, Higher% = 97, Mean% = 94

Alligation: (97-94):(94-90) = 3:4

Total = 21 litres. First solution (90%) = 3/(3+4) × 21 = 3/7 × 21 = 9 litres

SHORTCUT METHOD:

Alligation gives ratio 3:4. Multiply by 3 (since 7 parts = 21L). First solution = 9L.

ANSWER: 9 litres

Q9. [2019 | Slot 3 | Off-Campus]

QUESTION:

A container has 30 litres of alcohol and is filled with water to make it 50 litres. Some litres are drawn out and replaced with water. The ratio of alcohol to water is now 3:5. How many litres were drawn out?

STEP-BY-STEP SOLUTION:

Initial: 30L alcohol + 20L water = 50L. Alcohol% = 60%

Final ratio 3:5 → Alcohol% = 3/8 = 37.5%

Using replacement formula: 30 × [(50-x)/50] = 50 × (3/8)

30(50-x)/50 = 18.75

(50-x) = 18.75 × 50/30 = 31.25

x = 50 - 31.25 = 18.75 litres

SHORTCUT METHOD:

Find final alcohol. Use: Final alcohol = Initial alcohol × (V-x)/V. Solve for x.

ANSWER: 18.75 litres

Q10. [2018 | Slot 2 | Off-Campus]

QUESTION:

A jar full of whisky contains 40% alcohol. A part of this whisky is replaced by another containing 19% alcohol and now the percentage of alcohol was found to be 26%. The quantity replaced is:

STEP-BY-STEP SOLUTION:

Initial = 40%, Replacing liquid = 19%, Final = 26%

Alligation: (40-26):(26-19) = 14:7 = 2:1

Fraction replaced = 1/(2+1) = 1/3

So 1/3 of the whisky was replaced

SHORTCUT METHOD:

Alligation on percentages: (40-26):(26-19) = 14:7 = 2:1. Replaced fraction = 1/3.

ANSWER: 1/3 of the total quantity

Q11. [2018 | Slot 1 | Off-Campus]

QUESTION:

How many litres of pure acid must be added to 20 litres of a 20% acid solution to make it 45% acid solution?

STEP-BY-STEP SOLUTION:

Current: 20L of 20% solution → acid = 4L

Let x litres pure acid (100%) added

(4+x)/(20+x) = 45/100 = 9/20

20(4+x) = 9(20+x)
80 + 20x = 180 + 9x
11x = 100 → x = 100/11 ≈ 9.09 litres

SHORTCUT METHOD:

Use alligation: Cheaper=20%, Dearer=100%, Mean=45%. Ratio = (100-45):(45-20) = 55:25 = 11:5. 5 parts = 20L, so 11 parts = 44L. But pure acid = 44-20 = ... verify with direct method.

ANSWER: 100/11 litres ≈ 9.09 litres

Q12. [2017 | Slot 2 | Off-Campus]

QUESTION:

A shopkeeper bought 80 kg of sugar for Rs.2720. He sold 70 kg at Rs.40 per kg. At what rate per kg should he sell the remaining sugar so that he gains 20% on the total?

STEP-BY-STEP SOLUTION:

CP total = 2720. Desired profit = 20%, so SP total = 2720 × 1.2 = 3264
SP from 70 kg = 70 × 40 = 2800
SP needed from remaining 10 kg = 3264 - 2800 = 464
Rate per kg for remaining = 464/10 = Rs.46.40/kg

SHORTCUT METHOD:

Total required SP = CP × 1.2. Subtract known SP. Divide balance by remaining quantity.

ANSWER: Rs.46.40 per kg

Q13. [2017 | Slot 3 | Off-Campus]

QUESTION:

In what ratio must water be mixed with milk to gain 20% by selling the mixture at cost price?

STEP-BY-STEP SOLUTION:

To gain 20%, the cheapened cost must be 1/1.2 = 5/6 of selling price
CP of milk = 1 unit. Water = 0.
If milk:water = m:w, then CP of mixture = m/(m+w)
Profit = w/m = 20/100 = 1/5
So water:milk = 1:5

SHORTCUT METHOD:

Profit% = (Water/Milk) × 100 = 20%. Water/Milk = 1/5. Water:Milk = 1:5.

ANSWER: Water : Milk = 1 : 5

Q14. [2016 | Slot 1 | Off-Campus]

QUESTION:

A vessel of 90 litres is filled with milk and water in ratio 5:4. How many litres of water should be added to make ratio 5:6?

STEP-BY-STEP SOLUTION:

Milk = (5/9) × 90 = 50L, Water = 40L

Milk stays at 50L. New ratio 5:6 means water = 6/5 × milk = 6/5 × 50 = 60L
Water to add = 60 - 40 = 20 litres

SHORTCUT METHOD:
Milk is unchanged. Find required water from new ratio and subtract existing water.

ANSWER: 20 litres

Q15. [2016 | Slot 2 | Off-Campus]

QUESTION:
A mixture contains wine and water in ratio 3:2. If 4 litres of water is added, the ratio becomes 3:4. Find the quantity of wine in mixture.

STEP-BY-STEP SOLUTION:
Let wine = 3x, water = 2x
After adding 4L water: 3x/(2x+4) = 3/4
12x = 3(2x+4) = 6x+12
6x = 12 → x = 2
Wine = 3x = 6 litres

SHORTCUT METHOD:
Wine is unchanged. 3x/(2x+4) = 3/4 → 12x = 6x+12 → x=2 → wine = 6L.

ANSWER: 6 litres

Q16. [2015 | Slot 1 | Off-Campus]

QUESTION:
A barrel contains a mixture of wine and water in ratio 3:1. How much fraction of the mixture must be drawn off and substituted by water so that the ratio becomes 1:1?

STEP-BY-STEP SOLUTION:
Initial wine fraction = 3/4. Target wine fraction = 1/2
After removing fraction f and replacing with water:
New wine = (3/4)(1-f) = 1/2
(1-f) = (1/2)/(3/4) = 2/3
f = 1 - 2/3 = 1/3

SHORTCUT METHOD:
Wine after replacement = Initial wine × (1-f). Set equal to target. Solve for f = 1/3.

ANSWER: 1/3 of the mixture must be drawn and replaced

Q17. [2015 | Slot 3 | Off-Campus]

QUESTION:
Equal quantities of three types of coffee are mixed. The price of first is twice that of second and thrice that of third. Find the average price if the price of the second type is Rs.120/kg.

STEP-BY-STEP SOLUTION:
Price of second = 120, First = 2×120 = 240, Third = 240/3 = 80

Equal quantities, so weighted average = simple average
Average = (240 + 120 + 80)/3 = 440/3 = Rs.146.67/kg

SHORTCUT METHOD:

Equal quantities → simple average. Prices: 240, 120, 80. Average = 440/3.

ANSWER: Rs.146.67 per kg

Q18. [2014 | Slot 2 | Off-Campus]

QUESTION:

A dishonest milkman sells milk at CP but mixes water. If milk:water = 7:3, find his profit percentage.

STEP-BY-STEP SOLUTION:

Water = 3 parts, Milk = 7 parts
Profit% = (Water/Milk) × 100
= (3/7) × 100 = 300/7 ≈ 42.86%

SHORTCUT METHOD:

Profit% = (Water/Milk) × 100 = (3/7) × 100 = 42.86%

ANSWER: 42.86% (or 300/7 %)

Q19. [2014 | Slot 1 | Off-Campus]

QUESTION:

Two vessels A and B contain spirit and water mixed in ratio 5:2 and 7:6 respectively. Find ratio in which these mixtures be mixed to obtain new mixture in vessel C with spirit:water = 8:5.

STEP-BY-STEP SOLUTION:

Spirit in A = 5/7, Spirit in B = 7/13, Spirit in C = 8/13
Use alligation on spirit fraction:
Cheaper (B) = 7/13, Dearer (A) = 5/7, Mean (C) = 8/13
= 5/7 - 8/13 = (65-56)/91 = 9/91
= 8/13 - 7/13 = 1/13
Ratio A:B = (1/13) : (9/91) = (7/91):(9/91) = 7:9

SHORTCUT METHOD:

Convert all ratios to fractions. Apply alligation on spirit fraction. Ratio = (Mean-B):(A-Mean).

ANSWER: A : B = 7 : 9

Q20. [2013 | Slot 2 | Off-Campus]

QUESTION:

A container has 3 litres of a solution containing 25% alcohol. If another 3 litres of pure alcohol is added to this solution, what is the strength of the resultant solution?

STEP-BY-STEP SOLUTION:

Alcohol in first solution = 25% of 3L = 0.75L
Pure alcohol added = 3L

Total alcohol = 0.75 + 3 = 3.75L
Total solution = 3 + 3 = 6L
Concentration = (3.75/6) × 100 = 62.5%

SHORTCUT METHOD:

Total alcohol / Total solution × 100. (0.75+3)/6 = 3.75/6 = 62.5%

ANSWER: 62.5%

Q21. [2013 | Slot 1 | Off-Campus]

QUESTION:

A jar contains milk and water in ratio 5:1. 12 litres of the mixture are taken out and replaced with 12 litres of water. If the resulting ratio is 1:1, find the initial quantity in the jar.

STEP-BY-STEP SOLUTION:

Let total = V litres. Initial milk = 5V/6
After removing 12L and adding 12L water:
Milk = 5V/6 - (5/6 × 12) = 5V/6 - 10
Total still = V (volume unchanged)
New ratio = 1:1, so milk = V/2
5V/6 - 10 = V/2
5V/6 - V/2 = 10
(5V - 3V)/6 = 10 → 2V/6 = 10 → V = 30 litres

SHORTCUT METHOD:

Set up equation: Milk after = V/2. Solve. Milk removed = (5/6) × 12 = 10L.

ANSWER: 30 litres

Q22. [2012 | Slot 3 | Off-Campus]

QUESTION:

In a mixture of 60 litres, the ratio of milk and water is 2:1. If this ratio is to be 1:2, then the quantity of water to be added is:

STEP-BY-STEP SOLUTION:

Milk = 40L, Water = 20L
Milk stays 40L. For ratio 1:2, water = 2 × milk = 80L
Water to add = 80 - 20 = 60 litres

SHORTCUT METHOD:

Milk unchanged at 40L. New ratio 1:2 means water = 80L. Add 60L water.

ANSWER: 60 litres

Q23. [2012 | Slot 1 | Off-Campus]

QUESTION:

Rice worth Rs.126 per kg and Rs.135 per kg are mixed with a third variety in ratio 1:1:2. If the mixture is worth Rs.153 per kg, the price of the third variety per kg will be:

STEP-BY-STEP SOLUTION:

Let price of third variety = x
Ratio 1:1:2, so total = 4 parts
Mean = (1×126 + 1×135 + 2×x)/4 = 153
126 + 135 + 2x = 612
2x = 612 - 261 = 351 → x = 175.5

SHORTCUT METHOD:

Weighted average. (126+135+2x)/4=153 → 261+2x=612 → x=175.5

ANSWER: Rs.175.50 per kg

Q24. [2011 | Slot 2 | Off-Campus]

QUESTION:

A solution contains 8% salt. After adding 20 litres of water, the concentration becomes 4%. Find the original volume of solution.

STEP-BY-STEP SOLUTION:

Let original volume = V litres
Salt = 8% of V = 0.08V (unchanged)
After adding 20L water: 0.08V/(V+20) = 4/100
0.08V × 100 = 4(V+20)
8V = 4V + 80 → 4V = 80 → V = 20 litres

SHORTCUT METHOD:

Salt mass unchanged. Set up equation with new concentration. Solve: V = 20L.

ANSWER: 20 litres

Q25. [2011 | Slot 1 | Off-Campus]

QUESTION:

The ratio of water and alcohol in two different containers is 2:3 and 4:1 respectively. In what ratio should the mixtures of the two containers be mixed together so that the new mixture has half water and half alcohol?

STEP-BY-STEP SOLUTION:

Alcohol in container 1 = 3/5, Alcohol in container 2 = 1/5, Desired = 1/2
Alligation on alcohol fraction:
Cheaper (C2) = 1/5, Dearer (C1) = 3/5, Mean = 1/2
Ratio = (3/5 - 1/2):(1/2 - 1/5)
= (6/10 - 5/10):(5/10 - 2/10)
= (1/10):(3/10) = 1:3

SHORTCUT METHOD:

Apply alligation on alcohol fraction. Ratio C2:C1 = (3/5-1/2):(1/2-1/5) = 1/10:3/10 = 1:3.

ANSWER: Container 1 : Container 2 = 3 : 1

Q26. [2010 | Slot 1 | Off-Campus]

QUESTION:

A mixture of sugar and water has a ratio of 5:1. After adding 35 litres of water, the ratio becomes 5:3. Find the initial quantity of water.

STEP-BY-STEP SOLUTION:

Sugar stays constant. Initially sugar = 5x, water = x

After adding 35L: 5x/(x+35) = 5/3

15x = 5x + 175 → 10x = 175 → x = 17.5 litres

Initial water = 17.5 litres

SHORTCUT METHOD:

Sugar unchanged = 5x. 5x/(x+35) = 5/3 → x = 17.5

ANSWER: 17.5 litres

Q27. [2010 | Slot 3 | Off-Campus]

QUESTION:

A mixture contains spirit and water in ratio 3:2. How many litres of water must be added to 20 litres of this mixture to make ratio 3:5?

STEP-BY-STEP SOLUTION:

Mixture = 20L. Spirit = (3/5)×20 = 12L, Water = 8L

Spirit constant at 12L. New ratio 3:5 → water = (5/3) × 12 = 20L

Water to add = 20 - 8 = 12 litres

SHORTCUT METHOD:

Spirit = 12L. For 3:5 ratio, water = 20L. Add 12L.

ANSWER: 12 litres

Q28. [2009 | Slot 2 | Off-Campus]

QUESTION:

A solution of 30 litres contains milk and water in ratio 7:3. How many litres of water is to be added to make ratio 3:7?

STEP-BY-STEP SOLUTION:

Milk = 21L, Water = 9L

Milk constant at 21L. For 3:7 ratio: milk/water = 3/7

21/water = 3/7 → water = 49L

Add = 49 - 9 = 40 litres

SHORTCUT METHOD:

Milk = 21L. For 3:7, water = (7/3)×21 = 49L. Add 49-9 = 40L.

ANSWER: 40 litres

Q29. [2009 | Slot 1 | Off-Campus]

QUESTION:

Alloy A contains 40% gold and alloy B contains 60% gold. In what ratio should the alloys be mixed so that the resultant alloy contains 45% gold?

STEP-BY-STEP SOLUTION:

Cheaper (A) = 40%, Dearer (B) = 60%, Mean = 45%
Alligation: (60-45):(45-40) = 15:5 = 3:1

SHORTCUT METHOD:

(60-45):(45-40) = 15:5 = 3:1. Mix A and B in 3:1 to get 45% gold.

ANSWER: A : B = 3 : 1

Q30. [2008 | Slot 1 | Off-Campus]

QUESTION:

A dealer mixes rice costing Rs.6.75/kg with rice costing Rs.8.00/kg in ratio 2:3. He is 20% on it. At what price per kg does he sell the mixture?

STEP-BY-STEP SOLUTION:

CP of mixture = (2×6.75 + 3×8.00)/(2+3)
= (13.50 + 24.00)/5 = 37.50/5 = Rs.7.50/kg
SP = 7.50 × 1.20 = Rs.9.00/kg

SHORTCUT METHOD:

CP = weighted average = Rs.7.50. SP = 7.50 × 1.2 = Rs.9.

ANSWER: Rs.9.00 per kg

Q31. [2007 | Slot 2 | Off-Campus]

QUESTION:

A vessel of 120 litres is filled with milk. From the vessel, 24 litres of milk is drawn out and replaced by water. What is the percentage of water in the final mixture?

STEP-BY-STEP SOLUTION:

Milk after replacement = 120 × (96/120) = 120 × (4/5) = 96L
Water = 120 - 96 = 24L
Water% = (24/120) × 100 = 20%

SHORTCUT METHOD:

Water% = (x/V) × 100 = (24/120) × 100 = 20%

ANSWER: 20%

Q32. [2007 | Slot 1 | Off-Campus]

QUESTION:

In what ratio must three varieties of rice costing Rs.2, Rs.3.50 and Rs.4 per kg be mixed to obtain mixture costing Rs.2.50 per kg? (Use pair-wise alligation)

STEP-BY-STEP SOLUTION:

Pair 1: Rs.2 and Rs.4 mixed to get Rs.2.5
Ratio: (4-2.5):(2.5-2) = 1.5:0.5 = 3:1
Pair 2: Rs.3.5 and Rs.4 mixed to get Rs.2.5? (Rs.3.5 > Mean, so use Rs.2 and Rs.3.5)
Pair 2: Rs.2 and Rs.3.5: Ratio = (3.5-2.5):(2.5-2) = 1:0.5 = 2:1
Combine: For Rs.2 → 3+2=5, Rs.3.5 → 1, Rs.4 → 1

Ratio of Rs.2 : Rs.3.5 : Rs.4 = 5:1:3 (one valid answer)

SHORTCUT METHOD:

Use pairwise alligation. Multiple answers possible. One valid ratio: 5:1:3.

ANSWER: 5 : 1 : 3 (one possible ratio)

Q33. [2006 | Slot 1 | Off-Campus]

QUESTION:

A can contains a mixture of two liquids A and B in ratio 7:5. When 9 litres of mixture is drawn off and can is filled with B, the ratio of A and B becomes 7:9. How many litres does can hold?

STEP-BY-STEP SOLUTION:

Let can hold V litres. Initial A = 7V/12

After drawing 9L and adding 9L of B:

A = 7V/12 - (7/12)×9 = 7V/12 - 21/4

New ratio: A/total = 7/16

7V/12 - 21/4 = 7V/16

Multiply all by 48: 28V - 252 = 21V

7V = 252 → V = 36 litres

SHORTCUT METHOD:

Set up equation: A remaining = 7V/12 - 7/12×9. Set equal to 7V/16. Solve.

ANSWER: 36 litres

Q34. [2005 | Slot 1 | Off-Campus]

QUESTION:

A milk vendor has 2 cans of milk. The first contains 25% water and the rest milk. The second contains 50% water. How much milk should he mix from each can so as to get 12 litres of milk such that the ratio of water to milk is 3:5?

STEP-BY-STEP SOLUTION:

Target: Water:Milk = 3:5 → Milk% = 5/8 = 62.5%

Can 1: Milk% = 75%, Can 2: Milk% = 50%

Alligation: (75-62.5):(62.5-50) = 12.5:12.5 = 1:1

Equal quantities. Each can contributes 12/2 = 6 litres

SHORTCUT METHOD:

Find desired milk%. Use alligation on milk%. Ratio = 1:1, so 6L from each.

ANSWER: 6 litres from each can

Q35. [2024 | Slot 1 | Off-Campus]

QUESTION:

A mixture of 45 litres has milk and water in ratio 5:4. How many litres of water must be added to make the ratio 5:7?

STEP-BY-STEP SOLUTION:

Milk = (5/9) × 45 = 25L, Water = 20L

Milk constant at 25L. For 5:7: water = (7/5) × 25 = 35L
Water to add = 35 - 20 = 15 litres

SHORTCUT METHOD:

Milk = 25L. New water = (7/5) × 25 = 35L. Add 35-20 = 15L.

ANSWER: 15 litres

13. COMMON MISTAKES STUDENTS MAKE

#	Common Mistake	How to Avoid It
1	Confusing which value is 'cheaper' and 'dearer' in alligation	Always check: Cheaper < Mean < Dearer. If mean is outside this range, the problem is impossible.
2	Forgetting to raise the fraction to the POWER n in repeated replacement	Write the formula first, then substitute. Never skip steps.
3	Changing milk quantity when adding/removing WATER only	Only water changes when water is added or removed. Milk stays constant unless mixture is drawn.
4	Using alligation when weighted average is needed (and vice versa)	Alligation: you KNOW mean and WANT ratio. Weighted avg: you KNOW ratio and WANT mean.
5	Calculating profit% as (water/total) instead of (water/milk)	Profit% = Water/Milk × 100, NOT water/total × 100
6	Mixing two containers directly without computing per-unit fractions	Always compute milk fraction (milk/total) for each container before applying alligation.
7	Rounding off intermediate values causing wrong final answer	Keep fractions exact throughout. Convert to decimal only at the final step.
8	Applying replacement formula to mixed-start containers incorrectly	Formula gives MILK remaining = Initial milk × (fraction)^n, not total milk.

14. QUICK REVISION SHEET — 5-MINUTE REVISION

⚡ READ THIS THE NIGHT BEFORE YOUR TCS NQT EXAM ⚡

KEY FORMULAS	KEY RULES

1. ALLIGATION:
`Ratio = (D-M):(M-C)`
2. REPLACEMENT (n times):
`V × [(V-x)/V]^n`
3. PROFIT BY ADULTERATION:
`Profit% = (Water/Milk) × 100`
4. WEIGHTED AVERAGE:
`(Q1C1+Q2C2)/(Q1+Q2)`
	✓ Alligation: C < Mean < D always
	✓ Adding water → Milk quantity unchanged
	✓ Drawing mixture → Both milk & water reduce
	✓ % treated same as price in alligation
	✓ Pure water price = 0 in alligation
	✓ Pure liquid % = 100 in concentration problems
	✓ 3-ingredient: use pair-wise alligation
	✓ Replacement: raise fraction to power n

EXAM-DAY MENTAL FRAMEWORK

Step 1: Read → Is this ALLIGATION (find ratio given mean) or WEIGHTED AVG (find mean given ratio)?

Step 2: Is there REPLACEMENT happening? If yes, use the (V-x)/V formula.

Step 3: Is there addition of only water/milk? Track each component separately.

Step 4: Is this a PROFIT BY ADULTERATION problem? Use Profit% = (Adulterant/Pure) × 100.

— END OF CHAPTER: MIXTURES AND ALLIGATION —